SIMATS ENGINEERING

C02-ASSESSMENT TOOL-3

COURSE CODE: CSA1610 **COURSE NAME: DWDM**

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Q1. Data Cleaning and Normalization in Healthcare Analytics

A. Missing Value Treatment

i. Identify Missing Values

Attribute	Missing Record
Age	P2
BMI	P3
Annual_Visits	P5

ii. Mean Imputation

Age

Available values

25, 42, 110, 36, 50, 29, 45

Mean

=(25+42+110+36+50+29+45)/7

=337/7

= **48.14**

So,

P2 Age = **48.14**

BMI

Available values

22.5, 24.8, 45.2, 29.5, 31.2, 21.8, 60

Mean

=(22.5+24.8+45.2+29.5+31.2+21.8+60)/7

=235/7

= **33.57**

So,

P3 BMI = **33.57**

Annual Visits

Available values

3,5,7,50,12,4,40

Mean

=(3+5+7+50+12+4+40)/7

=121/7

= **17.29**

So,

P5 Annual_Visits = **17.29**

Median Imputation

Age

Sorted

25,29,36,42,45,50,110

Median = **42**

P2 Age = **42**

BMI

Sorted

21.8,22.5,24.8,29.5,31.2,45.2,60

Median = **29.5**

P3 BMI = **29.5**

Annual Visits

Sorted

3,4,5,7,12,40,50

Median = **7**

P5 Annual_Visits = **7**

Comparison

Attribute	**Mean**	**Median**
Age	48.14	42
BMI	33.57	29.5
Annual Visits	17.29	7

Justification

Median imputation is more appropriate because the dataset contains extreme values such as:

- Age = 110
- Annual Visits = 50
- BMI = 60

These outliers increase the mean, whereas the median remains unaffected. Therefore, **median imputation provides more reliable values**.

B. Outlier Detection

i. Z-Score Method

Formula

$$Z = \frac{x - \mu}{\sigma}$$

Age

Mean = 48.14

Standard deviation ≈ 28.09

Patient	**Age**	**Z-score**

P1	25	-0.82
P2	42	-0.22
P3	42	-0.22
P4	110	2.20
P5	36	-0.43
P6	50	0.07
P7	29	-0.68
P8	45	-0.11

Since none exceed ±3,

No statistical outlier, although **110 is an extreme observation**.

Annual Visits

Mean = 17.29

Standard deviation ≈18.54

Patient	Visits	Z-score
P1	3	-0.77
P2	5	-0.66
P3	7	-0.56
P4	50	1.77
P5	7	-0.56
P6	12	-0.29
P7	4	-0.72
P8	40	1.22

Again,

No Z-score exceeds ±3.

However,

50 visits is an unusually high observation.

ii. Extreme Records

- **P4 (Age =110)**
- **P4 (Annual Visits =50)**
- **P8 (Annual Visits =40)**

iii. Treatment

Instead of removing these records, **capping (Winsorization)** is preferred because:

- Healthcare data naturally contains elderly patients.
- Patients with chronic diseases may visit hospitals frequently.
- Removing such records may eliminate important clinical information.

Therefore,

Capping is the better approach.

C. Categorical Standardization

i. Inconsistent Gender Values

Original values

- M
- Male
- male
- F
- Female
- female
- FEMALE

ii. Standardized Values

Original	**Standard**

M	Male
Male	Male
male	Male
F	Female
Female	Female
female	Female
FEMALE	Female

Final Dataset

Patient	**Gender**
P1	Male
P2	Male
P3	Female
P4	Female
P5	Female
P6	Female
P7	Male
P8	Female

iii. Importance

Categorical standardization is important because:

- Removes duplicate categories.
- Improves data consistency.
- Prevents machine-learning algorithms from treating "Female" and "female" as different categories.
- Improves prediction accuracy.

D. Normalization

Formula

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Minimum BMI = **21.8**

Maximum BMI = **60**

Range = **38.2**

Normalized BMI

Patient	**BMI**	**Normalized BMI**
P1	22.5	0.02
P2	24.8	0.08
P3	29.5	0.20
P4	45.2	0.61
P5	29.5	0.20
P6	31.2	0.25
P7	21.8	0.00
P8	60	1.00

Effect of Normalization

Min-Max normalization scales all BMI values to the range **0–1**.

Advantages:

- Prevents attributes with large values from dominating smaller ones.
- Improves convergence speed of machine learning algorithms.
- Enhances model accuracy.
- Essential for algorithms such as KNN, Neural Networks, and Support Vector Machines.

Final Conclusion

The healthcare dataset was successfully preprocessed by:

- Identifying and imputing missing values.
- Detecting and treating outliers.
- Standardizing inconsistent categorical values.
- Normalizing BMI values.

These preprocessing techniques improve data quality, reduce bias, and enhance the performance and reliability of predictive healthcare models.

Q2. Covariance and Feature Selection in Banking Analytics

A. Covariance Analysis

i. Calculate the Covariance between Income and Loan Amount

Given Dataset

Customer	Income (₹K)	Loan Amount (₹K)
B1	35	100
B2	50	150
B3	70	200
B4	100	300
B5	42	120
B6	65	180
B7	130	350
B8	85	250

Step 1: Calculate Mean

Mean Income

$$\bar{X} = \frac{35 + 50 + 70 + 100 + 42 + 65 + 130 + 85}{8} = 72.125$$

Mean Loan Amount

$$\bar{Y} = \frac{100 + 150 + 200 + 300 + 120 + 180 + 350 + 250}{8} = 206.25$$

Step 2: Compute Covariance Table

Customer	Income (X)	Loan (Y)	X−X̄	Y−Ȳ	(X−X̄)(Y−Ȳ)
B1	35	100	-37.125	-106.25	3944.53
B2	50	150	-22.125	-56.25	1244.53
B3	70	200	-2.125	-6.25	13.28
B4	100	300	27.875	93.75	2613.28
B5	42	120	-30.125	-86.25	2598.28
B6	65	180	-7.125	-26.25	187.03
B7	130	350	57.875	143.75	8319.53
B8	85	250	12.875	43.75	563.28

Total

$$\sum (X - \bar{X})(Y - \bar{Y}) = 19483.75$$

Sample Covariance

$$Cov(X, Y) = \frac{19483.75}{7}$$
$$\boxed{Covariance \approx 2783.39}$$

ii. Relationship

The covariance value is **positive**.

This indicates that:

- Higher income generally corresponds to higher loan amounts.
- Lower income customers usually take smaller loans.

iii. Interpretation

A **high positive covariance** means:

- Income and Loan Amount increase together.
- Customers earning more tend to qualify for larger loans.
- These variables have a strong linear relationship.

B. Feature Relationship Analysis

i. Highly Correlated Feature Pairs

The following attributes are likely to be strongly correlated:

Feature 1	**Feature 2**	**Reason**
Income	Loan Amount	Higher income customers receive larger loans.
Income	Savings	Customers with higher income generally save more.
Loan Amount	Monthly EMI	Higher loans lead to higher EMI payments.
Age	Income	Income generally increases with age and experience.
Savings	Risk	Customers with greater savings usually have lower financial risk.

ii. Why Age, Income, Savings, and Loan Amount May Be Redundant

These variables contain overlapping information.

For example:

- Older customers often have higher salaries.
- Higher salaries lead to greater savings.
- Higher income customers are eligible for larger loans.

Since these features move together, they provide similar information to the prediction model.

This redundancy is called **multicollinearity**.

iii. Features That Could Be Removed

Possible removable features include:

- **Age**
- **Savings**
- **Monthly EMI**

These features overlap with Income and Loan Amount and may not significantly improve prediction accuracy.

C. Feature Selection

i. Selected Features

The optimal feature subset is:

- Income
- Credit Score
- Loan Amount
- Monthly EMI

Target Variable:

- Risk

ii. Justification

Income

Shows customer's repayment capability.

Credit Score

Directly measures creditworthiness.

Loan Amount

Represents financial burden.

Monthly EMI

Shows repayment commitment.

These four features are highly relevant for predicting customer risk.

iii. How Feature Selection Reduces Computational Cost

Feature selection:

- Removes unnecessary variables.
- Reduces memory usage.
- Speeds up model training.
- Reduces overfitting.
- Improves model interpretability.
- Increases prediction efficiency.

D. Standardization (Z-score Normalization)

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Mean Values

Feature	Mean
Age	38.88
Income	72.13
Loan Amount	206.25
Credit Score	722.5
Monthly EMI	9750
Savings	47.5

Standardized Dataset (Approximate)

Customer	Age	Income	Loan	Credit Score	EMI	Savings
B1	-1.30	-1.07	-1.15	-1.18	-1.10	-1.01
B2	-0.73	-0.57	-0.64	-0.74	-0.66	-0.65

B3	0.05	-0.07	-0.07	-0.04	-0.22	-0.11
B4	0.67	0.93	1.06	0.84	0.88	0.80
B5	-0.99	-0.87	-0.92	-1.03	-0.99	-0.83
B6	-0.16	-0.24	-0.30	-0.19	-0.33	-0.29
B7	1.46	1.93	1.63	1.42	1.54	1.52
B8	0.46	0.43	0.49	0.55	0.88	0.58

(Values rounded to two decimal places.)

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ii. Why Credit Score and Income Cannot Be Directly Compared Without Scaling

Credit Score and Income have different units and ranges.

- Income is measured in thousands of rupees (₹K).
- Credit Score ranges roughly between 300 and 900.

Without scaling:

- Income values are numerically much larger.
- Machine learning algorithms may assign more importance to Income simply because of its larger scale.

Z-score normalization transforms all features to the same scale (mean = 0, standard deviation = 1), ensuring that each feature contributes fairly to the model.

Final Conclusion

The banking dataset analysis shows that **Income** and **Loan Amount** have a strong positive covariance, indicating they increase together. Several attributes, such as **Age**, **Savings**, and **Monthly EMI**, contain redundant information and can be removed to reduce model complexity. Selecting the most relevant features and applying **Z-score standardization** improves model efficiency, reduces computational cost, and enhances the accuracy of customer risk prediction.

Q3. Discretization of Sales Data in Retail Analytics

A. Equal-Width Binning

i. Divide Monthly_Spend into 4 Equal-Width Bins

Given Dataset

Customer_ID	**Monthly_Spend (₹)**
C1	500
C2	750
C3	1000
C4	1500
C5	2000
C6	2500
C7	3000
C8	4000
C9	5000
C10	6500

ii. Calculate the Bin Width

Minimum Monthly Spend = **500**

Maximum Monthly Spend = **6500**

Number of Bins = **4**

Formula:

$$\text{Bin Width} = \frac{\text{Maximum} - \text{Minimum}}{\text{Number of Bins}}$$
$$= \frac{6500 - 500}{4} = \frac{6000}{4} = 1500$$

Bin Width = ₹1500

iii. Define the Intervals and Assign Customers

Bin	**Interval (₹)**	**Customers**
Bin 1	500 – <2000	C1, C2, C3, C4

Bin 2	2000 – <3500	C5, C6, C7
Bin 3	3500 – <5000	C8
Bin 4	5000 – 6500	C9, C10

Equal-Width Binning Result

Customer	Monthly Spend (₹)	Assigned Bin
C1	500	Bin 1
C2	750	Bin 1
C3	1000	Bin 1
C4	1500	Bin 1
C5	2000	Bin 2
C6	2500	Bin 2
C7	3000	Bin 2
C8	4000	Bin 3
C9	5000	Bin 4
C10	6500	Bin 4

B. Equal-Frequency Binning

i. Divide into 4 Bins with Approximately Equal Records

There are **10 customers**.

Each bin should contain approximately:

$$10/4 = 2.5$$

So, we create bins containing **2–3 customers**.

Equal-Frequency Bins

Bin	Customers	Monthly Spend (₹)

Bin 1	C1, C2, C3	500, 750, 1000
Bin 2	C4, C5, C6	1500, 2000, 2500
Bin 3	C7, C8	3000, 4000
Bin 4	C9, C10	5000, 6500

ii. Customer Assignment

Customer	Monthly Spend (₹)	Equal-Frequency Bin
C1	500	Bin 1
C2	750	Bin 1
C3	1000	Bin 1
C4	1500	Bin 2
C5	2000	Bin 2
C6	2500	Bin 2
C7	3000	Bin 3
C8	4000	Bin 3
C9	5000	Bin 4
C10	6500	Bin 4

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iii. Comparison

Equal-Width Binning	Equal-Frequency Binning
Uses fixed interval widths.	Uses an equal number of records in each bin.
Some bins may contain many records while others contain few.	Each bin has nearly the same number of records.
Simple to interpret.	Better balances the data distribution.
Sensitive to skewed data.	Less affected by skewed distributions.

Customer Categories

Customer	Monthly Spend (₹)	Category
C1	500	Low
C2	750	Low
C3	1000	Low
C4	1500	Medium
C5	2000	Medium
C6	2500	Medium
C7	3000	High
C8	4000	High
C9	5000	Very High
C10	6500	Very High

D. Interpretation

i. How Discretization Improves Decision-Making

Discretization converts continuous numerical values into meaningful categories, making the data easier to understand and analyze.

Advantages:

- Simplifies complex numerical data.
- Makes reports easier for managers to interpret.
- Helps identify customer segments quickly.
- Improves visualization and business reporting.
- Useful for rule-based classification and decision-making.

Example:

Instead of saying a customer spends **₹5,000**, managers can classify them as a **"Very High Spending Customer"**, making marketing strategies easier to plan.

ii. Disadvantage of Discretization

One disadvantage is the **loss of detailed information**.

For example:

- Customers spending **₹5,000** and **₹6,500** are both placed in the **Very High** category, even though their spending differs significantly.

This reduction in precision may slightly decrease the accuracy of machine learning models because some numerical information is lost.

Final Conclusion

The retail sales dataset was successfully discretized using both **Equal-Width** and **Equal-Frequency** binning techniques. A concept hierarchy was created to categorize customers into **Low**, **Medium**, **High**, and **Very High** spending groups. Discretization improves data interpretation, customer segmentation, and managerial decision-making, although it may result in some loss of detailed numerical information. This preprocessing step is particularly useful in retail analytics for targeted marketing and customer relationship management.

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C10	6500	Bin 4

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The retail sales dataset was successfully discretized using both **Equal-Width** and **Equal-Frequency** binning techniques. A concept hierarchy was created to categorize customers into **Low**, **Medium**, **High**, and **Very High** spending groups. Discretization improves data interpretation, customer segmentation, and managerial decision-making, although it may result in some loss of detailed numerical information. This preprocessing step is particularly useful in retail analytics for targeted marketing and customer relationship management.

Q4. Data Transformation in Manufacturing Quality Analytics

A. Min-Max Normalization

Given Dataset

Machine_ID	Temperature (°C)	Operating_Hours	Vibration
M1	55	100	2.1
M2	60	120	2.5
M3	62	140	2.8
M4	65	160	3.0
M5	70	180	3.2
M6	75	200	3.8
M7	80	220	4.1
M8	120	250	8.5

i. Normalize Temperature into the Range [0,1]

Formula

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Where:

- Minimum Temperature = **55°C**
- Maximum Temperature = **120°C**

Range:

$$120 - 55 = 65$$

ii. Normalized Temperature Values

Machine	Temperature (°C)	Calculation	Normalized Value
M1	55	(55−55)/65	0.00
M2	60	(60−55)/65	0.08
M3	62	(62−55)/65	0.11
M4	65	(65−55)/65	0.15
M5	70	(70−55)/65	0.23
M6	75	(75−55)/65	0.31
M7	80	(80−55)/65	0.38
M8	120	(120−55)/65	1.00

iii. Observation with Maximum Normalized Value

The highest normalized value is:

Machine	**Temperature**	**Normalized Value**
M8	**120°C**	**1.00**

Therefore, **Machine M8** obtains the maximum normalized value.

B. Z-Score Normalization

Formula

$$Z = \frac{X - \mu}{\sigma}$$

i. Mean Temperature

$$\mu = \frac{55 + 60 + 62 + 65 + 70 + 75 + 80 + 120}{8}$$
$$= \frac{587}{8}$$
$$= 73.38^{\circ}C$$

Mean Temperature = 73.38°C

ii. Standard Deviation

Using the standard deviation formula,

$$\sigma \approx 20.22^{\circ}C$$

iii. Z-Score for Each Temperature

Machine	Temperature (°C)	Z-Score
M1	55	-0.91
M2	60	-0.66
M3	62	-0.56
M4	65	-0.41
M5	70	-0.17
M6	75	0.08
M7	80	0.33
M8	120	2.31

(Rounded to two decimal places.)

C. Outlier Analysis

i. Is 120°C an Extreme Value?

According to the Z-score method:

- If **|Z| > 3**, the observation is considered an outlier.

For **120°C**:

$$Z = 2.31$$

Since:

$$2.31 < 3$$

120°C is not a statistical outlier, but it is an **extreme observation** because it is much higher than the other temperature values.

ii. Effect on Mean-Based Normalization

The temperature value of **120°C** increases the maximum value in the dataset.

As a result:

- Other normalized temperature values become compressed toward **0**.
- The differences among normal machines become less noticeable.
- Min-Max normalization becomes more sensitive to the extreme value.

iii. Appropriate Treatment Method

Instead of removing the value, **capping (Winsorization)** is recommended.

Justification:

- A high temperature may indicate a genuine machine fault.
- Removing the observation could eliminate important maintenance information.
- Capping reduces the influence of the extreme value while preserving the record.

D. Comparison of Min-Max and Z-Score Normalization

Aspect	**Min-Max Normalization**	**Z-Score Normalization**
Range	0 to 1	No fixed range
Formula	(X−Min)/(Max−Min)	(X−Mean)/Standard Deviation

Effect of Outliers	Highly sensitive	Less sensitive
Interpretation	Easy to understand	Shows distance from the mean
Best Used For	Neural Networks, KNN	Regression, Clustering, PCA
Distribution	Preserves relative order	Centers data around mean (0)

Sensitivity to Extreme Observations

- **Min-Max Normalization** is highly affected by extreme values because it depends on the minimum and maximum values. A single unusually high value (such as 120°C) can compress the remaining normalized values into a smaller range.
- **Z-Score Normalization** is less sensitive because it uses the mean and standard deviation. Although extreme values influence these statistics, their impact is generally less severe than in Min-Max normalization.

Final Conclusion

The manufacturing dataset was successfully transformed using both **Min-Max** and **Z-Score normalization**. Temperature values were scaled to improve comparability, and the extreme reading of **120°C** was analyzed using the Z-score method. While it is not a statistical outlier, it significantly affects Min-Max normalization. Applying appropriate preprocessing techniques such as normalization and capping improves data quality, supports accurate quality analysis, and enhances the performance of machine learning models used for manufacturing quality monitoring.

Q5. Missing Value Imputation and Data Transformation in Student Analytics

A. Missing Value Analysis

i. Identify Missing Observations

The dataset contains the following missing values:

Student_ID	**Attribute**	**Missing Value**
S2	Internal_Mark	NaN
S3	Attendance (%)	NaN
S5	Study_Hours	NaN

S6	Assignment_Score	NaN

ii. Apply Mean Imputation and Median Imputation

1. Study_Hours

Available values:

2, 3, 4, 5, 7, 8, 12

Mean

$$\frac{2+3+4+5+7+8+12}{7} = \frac{41}{7} = 5.86$$

Mean Imputation:

S5 = 5.86 hours

Median

Sorted values:

2, 3, 4, 5, 7, 8, 12

Median = **5**

Median Imputation:

S5 = 5 hours

2. Internal_Mark

Available values:

55, 68, 75, 80, 88, 92, 98

Mean

$$\frac{55+68+75+80+88+92+98}{7} = \frac{556}{7} = 79.43$$

Mean Imputation:

S2 = 79.43

Median

Sorted values:

55, 68, 75, 80, 88, 92, 98

Median = **80**

Median Imputation:

S2 = 80

3. Attendance (%)

Available values:

70, 75, 85, 90, 92, 95, 98

Mean

$$\frac{70+75+85+90+92+95+98}{7} = \frac{605}{7} = 86.43$$

Mean Imputation:

S3 = 86.43

Median

Sorted values:

70, 75, 85, 90, 92, 95, 98

Median = **90**

Median Imputation:

S3 = 90

4. Assignment_Score

Available values:

60, 65, 72, 78, 82, 90, 95

Mean

$$\frac{60 + 65 + 72 + 78 + 82 + 90 + 95}{7} = \frac{542}{7} = 77.43$$

Mean Imputation:

S6 = 77.43

Median

Sorted values:

60, 65, 72, 78, 82, 90, 95

Median = **78**

Median Imputation:

S6 = 78

iii. Suitable Method for Study_Hours

Median imputation is more suitable because:

- The dataset contains a relatively high value (**12 hours**) compared to the other study hours.
- This value increases the mean.
- The median is less affected by extreme values and better represents the central tendency.

Therefore, **median imputation is preferred for Study_Hours**.

B. Data Transformation

i. Apply Min-Max Normalization to Internal_Mark

Using the **mean-imputed value (79.43)** for S2.

Formula

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Minimum Internal Mark = **55**

Maximum Internal Mark = **98**

Range = **43**

Normalized Internal Marks

Student	Internal Mark	Normalized Value
S1	55	0.00
S2	79.43	0.57
S3	68	0.30
S4	75	0.47
S5	80	0.58
S6	88	0.77
S7	92	0.86
S8	98	1.00

ii. Apply Z-Score Normalization to Assignment_Score

Using **mean-imputed value (77.43)** for S6.

Mean Assignment Score

$$77.43$$

Standard Deviation

Approximately

$$11.74$$

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Z-Score Values

Student	Assignment Score	Z-Score
S1	60	-1.48
S2	65	-1.06
S3	72	-0.46
S4	78	0.05
S5	82	0.39
S6	77.43	0.00
S7	90	1.07
S8	95	1.50

(Values rounded to two decimal places.)

iii. Compare the Transformed Results

Min-Max Normalization	Z-Score Normalization
Scales values between 0 and 1.	Centers data around a mean of 0 with a standard deviation of 1.
Easy to interpret.	Useful for identifying deviations from the mean.
Sensitive to outliers.	Less sensitive to outliers.
Commonly used in Neural Networks and KNN.	Commonly used in Regression, PCA, and Clustering.

C. Discretization

Divide Study_Hours into Three Categories

Suitable thresholds:

- **Low:** 0 – 3 hours
- **Medium:** 4 – 7 hours
- **High:** More than 7 hours

Using the **median-imputed value (5 hours)** for S5:

Student	Study Hours	Category

S1	2	Low
S2	3	Low
S3	4	Medium
S4	5	Medium
S5	5	Medium
S6	7	Medium
S7	8	High
S8	12	High

D. Interpretation

Data preprocessing plays an important role in improving the quality and reliability of a student-performance prediction model.

Benefits of Data Preprocessing:

- Missing value imputation ensures that incomplete records can still be used without discarding valuable information.
- Normalization scales numerical features to a common range, preventing attributes with larger values from dominating the model.
- Discretization converts continuous values into meaningful categories, making the data easier to interpret and analyze.
- Proper preprocessing reduces noise and inconsistencies, leading to better model accuracy and more reliable predictions.
- It also improves the interpretability of results, allowing educators to identify students who may need additional academic support.